EX NO:1	Butterworth IIR Filters
DATE:	1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

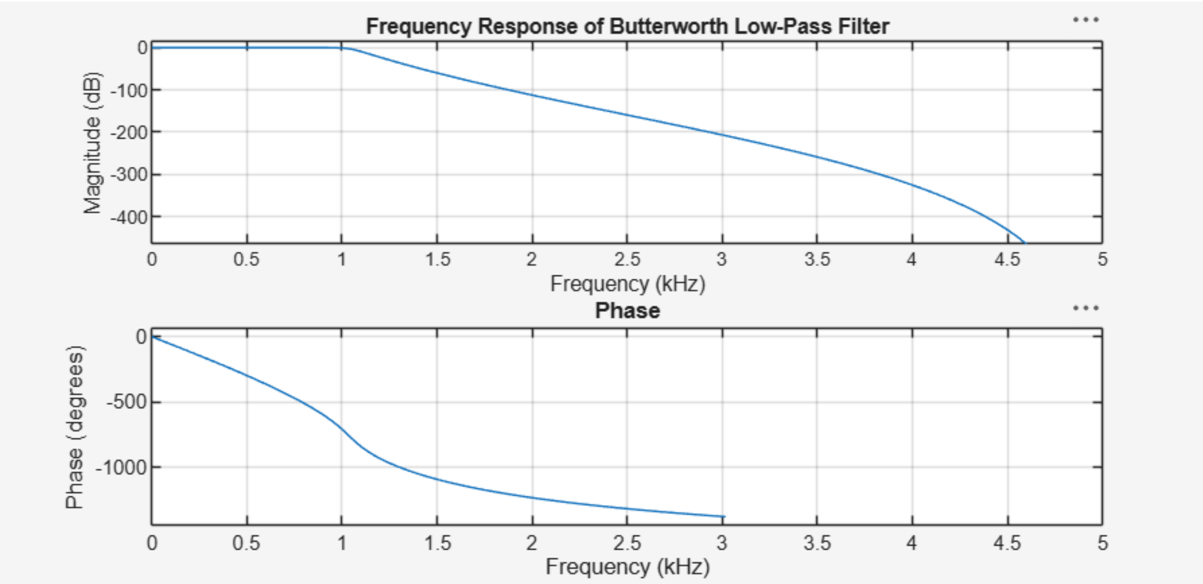
EXP 1

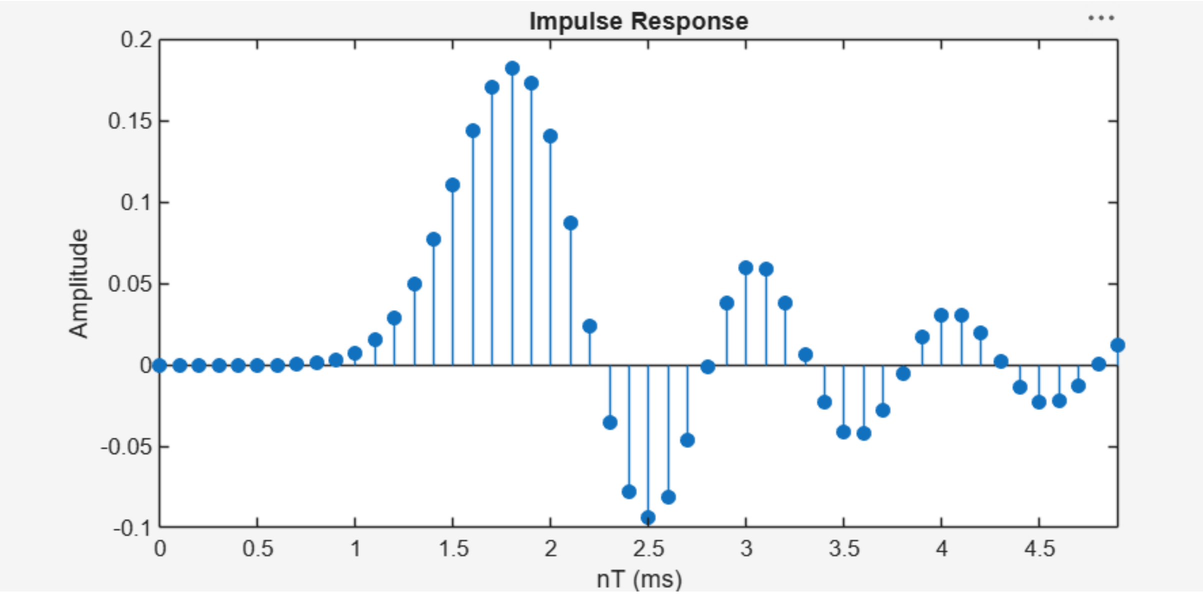
Code:

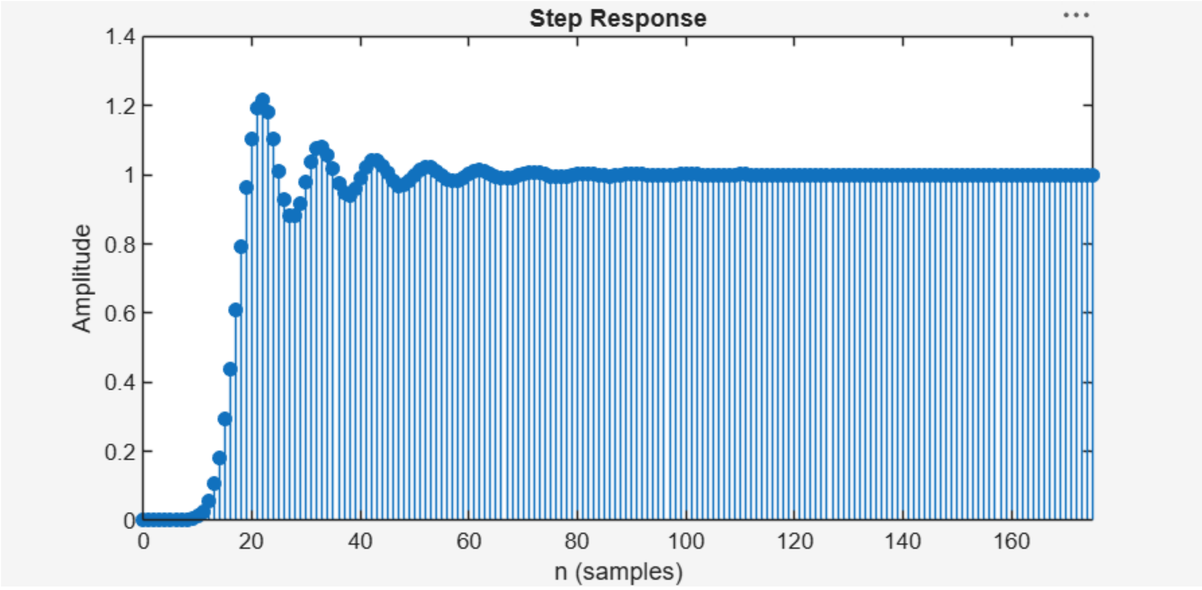
```
clc;
clear;
close all;

%-------------------------------------------------------
% EX NO : 1
% TITLE : Design and Analysis of a Butterworth IIR
%         Low-Pass Filter Using MATLAB
%-------------------------------------------------------

% Specifications
Fs  = 10000;      % Sampling Frequency (Hz)
Fp  = 1000;       % Passband Frequency (Hz)
Fst = 1500;       % Stopband Frequency (Hz)
Rp  = 1;          % Passband Ripple (dB)
Rs  = 60;         % Stopband Attenuation (dB)

% Determine Minimum Filter Order
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth Low-Pass Filter
[b, a] = butter(N, Wn, 'low');

%-------------------------------------------------------
% Frequency Response
%-------------------------------------------------------
figure;
freqz(b, a, 1024, Fs);
title('Frequency Response of Butterworth Low-Pass Filter');

%-------------------------------------------------------
% Impulse Response
%-------------------------------------------------------
figure;
impz(b, a, 50, Fs);
title('Impulse Response');

%-------------------------------------------------------
% Step Response
%-------------------------------------------------------
figure;
stepz(b, a);
title('Step Response');

%-------------------------------------------------------
% Pole-Zero Plot
%-------------------------------------------------------
```

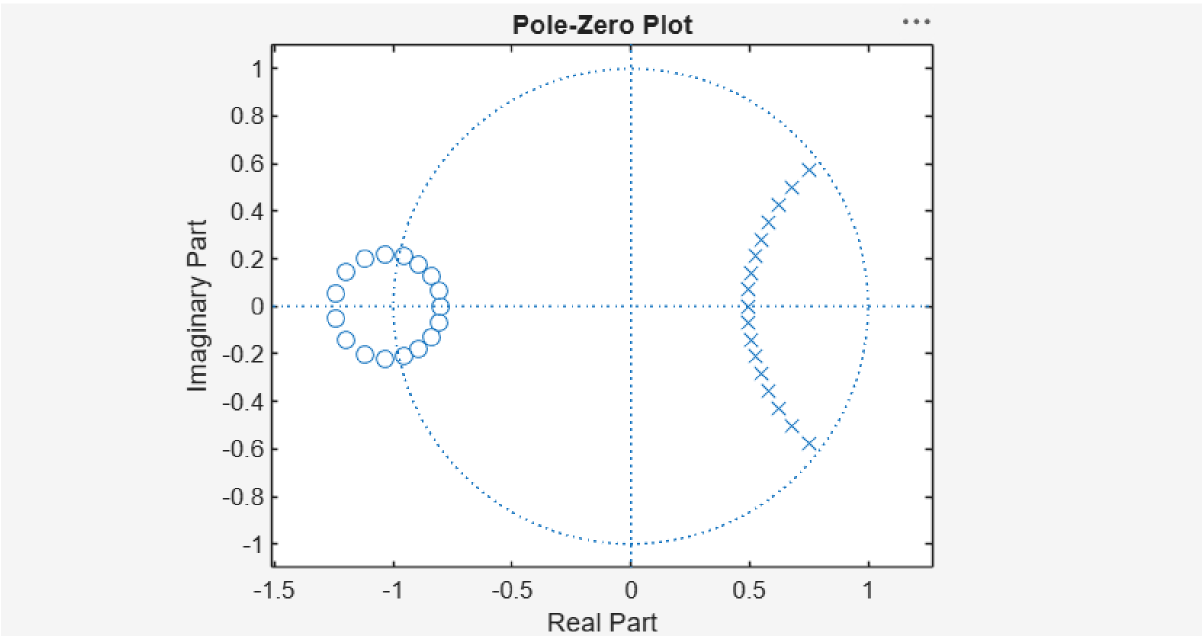
```
figure;
zplane(b, a);
title('Pole-Zero Plot');

%-------------------------------------------------------
% Display Results
%-------------------------------------------------------
fprintf('--------------------------------------\n');
fprintf('Butterworth Low-Pass Filter\n');
fprintf('--------------------------------------\n');
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);
fprintf('Sampling Frequency = %d Hz\n', Fs);
fprintf('Passband Frequency = %d Hz\n', Fp);
fprintf('Stopband Frequency = %d Hz\n', Fst);
fprintf('--------------------------------------\n');
```

Output;









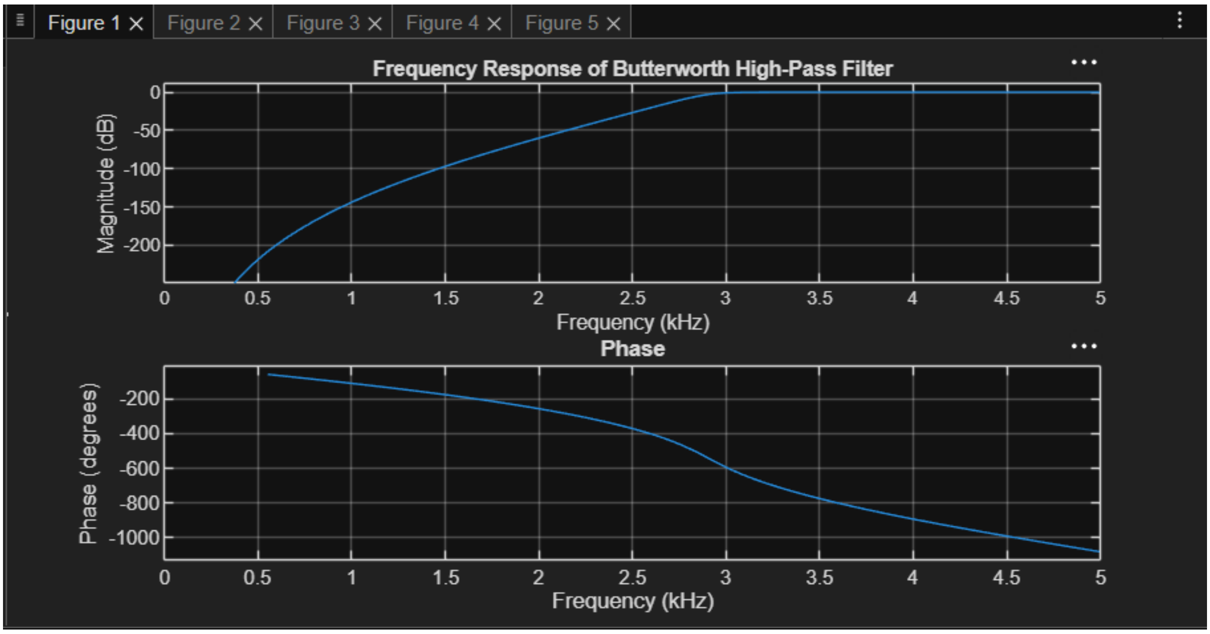

```
Passband Frequency = 1000 Hz
Stopband Frequency = 1500 Hz
--------------------------------------
```

EX NO:1	Butterworth IIR Filters
DATE:	1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

```
clc;
clear;
close all;

%-------------------------------------------------------
% EX NO : 1
% TITLE : Design and Analysis of a Butterworth IIR
%         High-Pass Filter Using MATLAB
%-------------------------------------------------------

% Filter Specifications
Fs  = 10000;      % Sampling Frequency (Hz)
Fp  = 3000;       % Passband Frequency (Hz)
Fst = 2000;       % Stopband Frequency (Hz)
Rp  = 1;          % Passband Ripple (dB)
Rs  = 60;         % Stopband Attenuation (dB)

% Determine Minimum Filter Order
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth High-Pass Filter
[b, a] = butter(N, Wn, 'high');

%-------------------------------------------------------
% Frequency Response
%-------------------------------------------------------
figure;
freqz(b, a, 1024, Fs);
title('Frequency Response of Butterworth High-Pass Filter');

%-------------------------------------------------------
% Impulse Response
%-------------------------------------------------------
figure;
[h, n] = impz(b, a, 50);
stem(n, h, 'filled');
grid on;
title('Impulse Response');
xlabel('Samples');
ylabel('Amplitude');

%-------------------------------------------------------
% Step Response
%-------------------------------------------------------
figure;
stepz(b, a);
title('Step Response');

%-------------------------------------------------------
% Group Delay
```

```
%-------------------------------------------------------
figure;
grpdelay(b, a);
title('Group Delay');

%-------------------------------------------------------
% Pole-Zero Plot
%-------------------------------------------------------
figure;
zplane(b, a);
title('Pole-Zero Plot');

%-------------------------------------------------------
% Display Results
%-------------------------------------------------------
fprintf('--------------------------------------\n');
fprintf('Butterworth High-Pass Filter\n');
fprintf('--------------------------------------\n');
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);
fprintf('Sampling Frequency = %d Hz\n', Fs);
fprintf('Passband Frequency = %d Hz\n', Fp);
fprintf('Stopband Frequency = %d Hz\n', Fst);
fprintf('--------------------------------------\n');
```

Output;



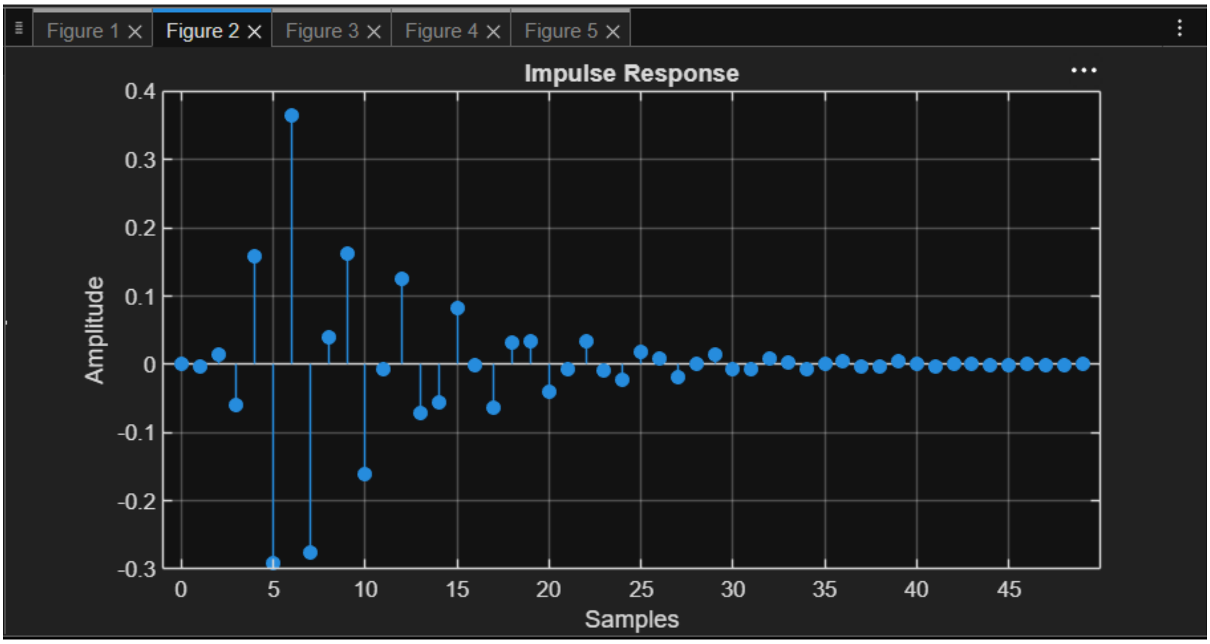


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Impulse Response
Amplitude
Samples
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
0
5
10
15
20
25
30
35
40
45

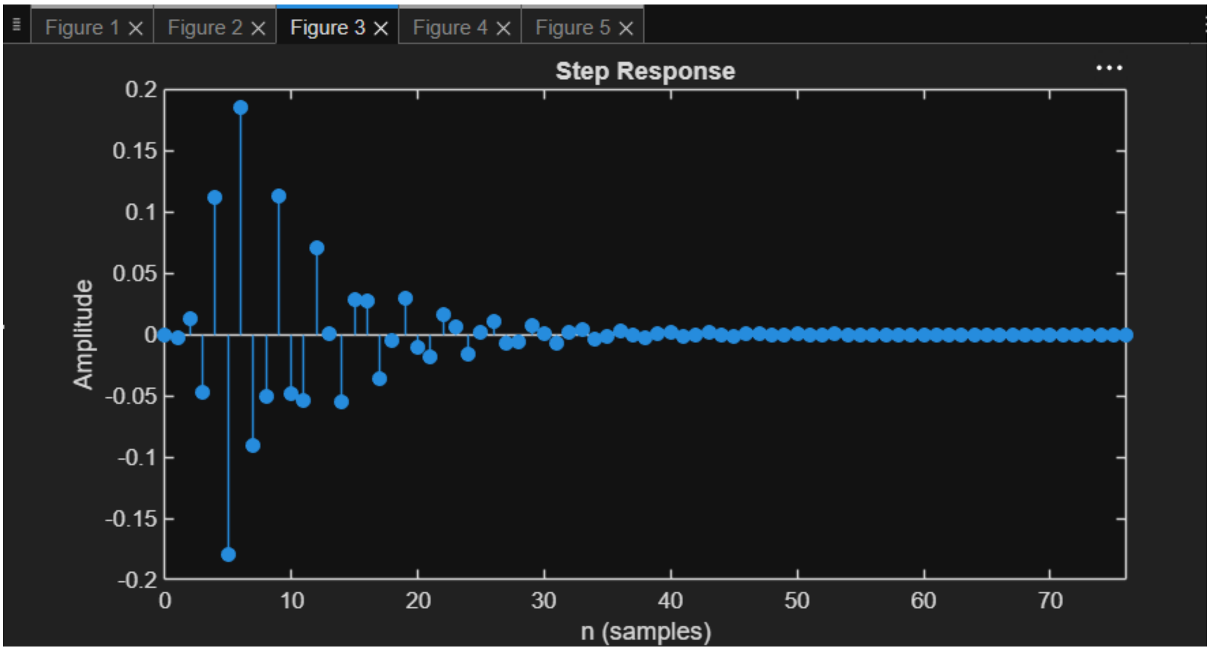


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Step Response
Amplitude
n (samples)
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
-0.2
0
10
20
30
40
50
60
70

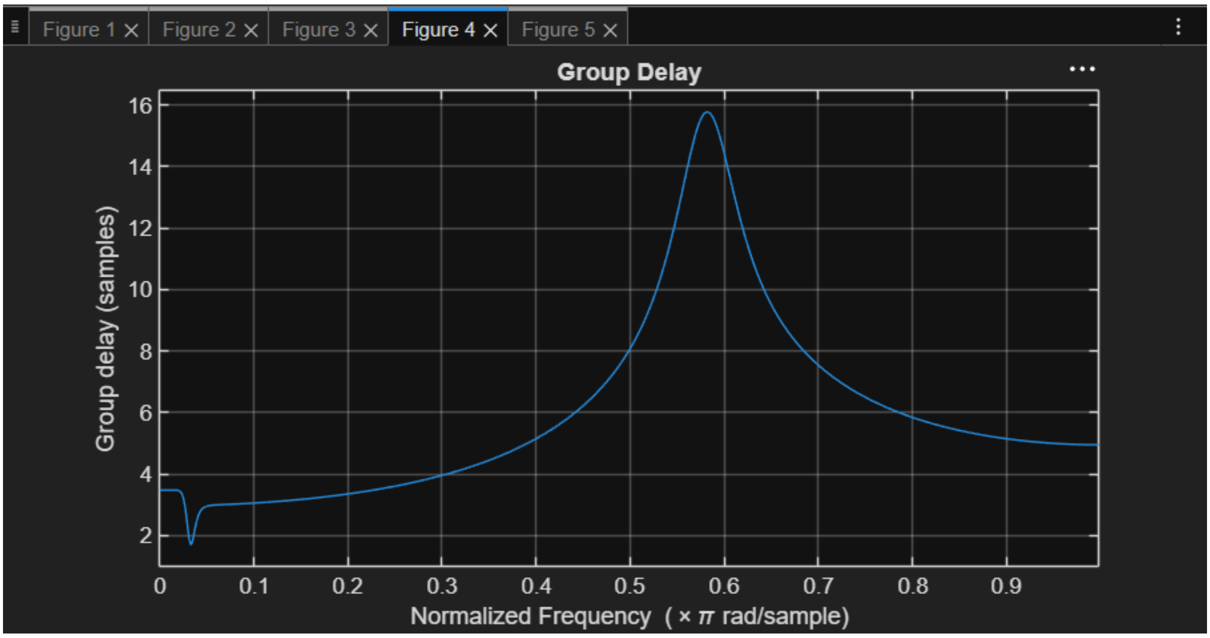


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Group Delay
Group delay (samples)
Normalized Frequency (× π rad/sample)

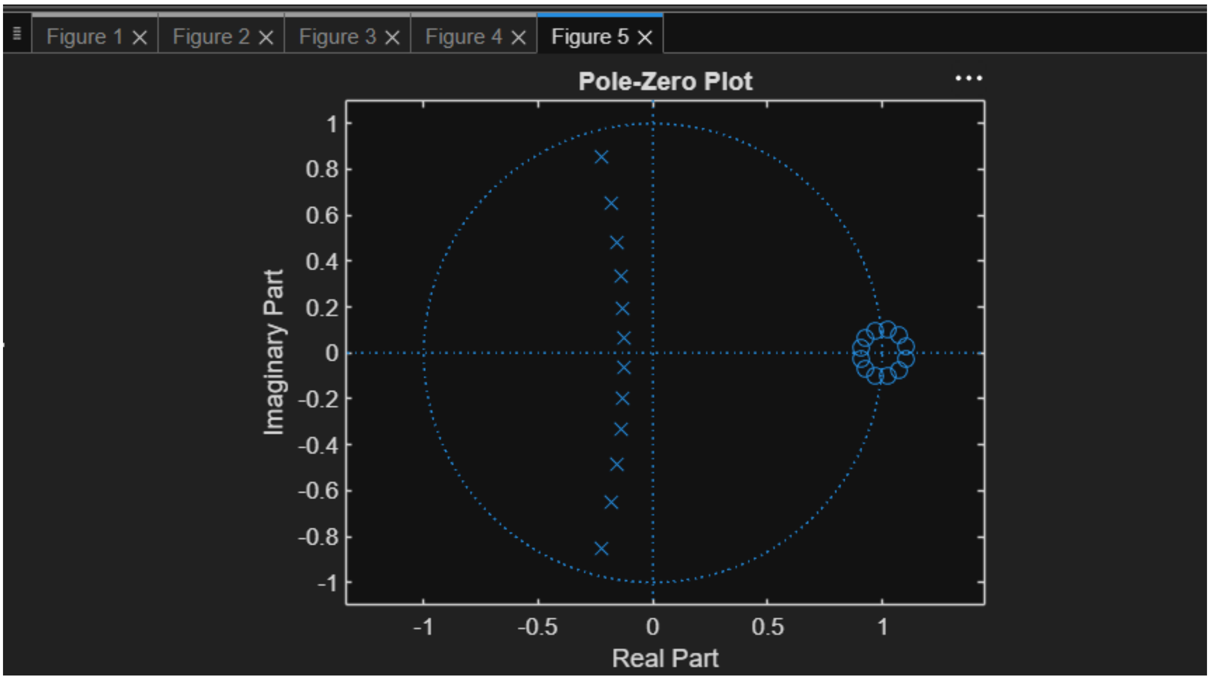


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Pole-Zero Plot
Imaginary Part
Real Part